Asian Journal of Psychiatry 113 (2025) 104707

Contents lists available at ScienceDirect

Asian Journal of Psychiatry

journal homepage: www.elsevier.com/locate/ajp

Can AI replace clinician-rated depression scales? The psychometric properties of HAMLET – Hamilton Large-Language-Model Evaluation Tool

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ARTICLE INFO

Keywords:
Large Language Models
Hamilton Depression Rating Scale
Automated Clinical Assessment
Scale Validation

ABSTRACT

Background: The Hamilton Depression Rating Scale (HAMD) is the gold standard for assessing depression but requires clinician administration, limiting its accessibility. Large language models (LLMs) offer potential for automated, valid assessments. We developed HAMLET (Hamilton Large-language-model Evaluation Tool), an interactive LLM-based tool designed to replicate the HAMD-17, and compared its results with those obtained by a psychiatrist.

Methods: HAMLET utilizes Qwen-Max in the temperature of 0.7 and a top-p value of 0.6, guided by structured prompts engineered via clinician-supervised tuning. 60 patients with Major Depressive Disorder completed: (1) HAMLET, (2) clinician-rated HAMD-17, and (3) PHQ-9. Agreement was assessed using Intraclass Correlation Coefficient (ICC), Bland-Altman plots, and Gwet's AC2. Correlations with HAMD and incremental validity beyond the PHQ-9 were evaluated.

Results: HAMLET demonstrated strong overall agreement with clinician-rated HAMD scores (ICC=0.911; 95 % CI: 0.855–0.946). It outperformed the PHQ-9 in correlating with HAMD scores ($r = 0.92$ vs 0.79; Steiger's $Z = 3.798$, $p < 0.001$) and demonstrated incremental validity (ΔR^2=0.252). Item-level agreement (Gwet's AC2) was > 0.60 for all items, though was lower for sensitive questions.

Conclusion: HAMLET is the first LLM framework to autonomously conduct HAMD-17 assessments demonstrating substantial agreement with clinician ratings. It combines the rigor of clinician-rated scales with the accessibility of self-report tools. This demonstrates the feasibility of LLMs for scalable, low-cost psychiatric assessment. Future work should address contextual limitations and explore multimodal integration.

1. Introduction

Since its introduction in the 1960s (Hamilton, 1960), the Hamilton Depression Rating Scale (HAMD) has long been regarded as the gold standard for assessing depression. It has been cited in over 90 % of antidepressant trials (Yin et al., 2019) and is recognized by the U.S. Food and Drug Administration (FDA) as a primary efficacy endpoint in antidepressant clinical trials (Bagby et al., 2004). However, the HAMD requires administration by experienced clinicians (O'Hara and Rehm, 1983), creating significant accessibility barriers. In clinical practice, most physicians rarely use rating scales (Demyttenaere and Jaspers, 2020), citing time constraints and concerns over validity.

Self-report scales, while more efficient and convenient, often lack the diagnostic rigor of clinician-administered tools. They are more susceptible to patient-introduced biases. Studies have found that clinician-rated scales consistently produce significantly higher effect sizes than self-report scales within the same study (Cuijpers et al., 2010).

Ecological Momentary Assessment (EMA) offers an alternative by enabling real-time symptom monitoring with reduced recall bias (Shiffman et al., 2008). However, EMA methods still require cross-validation against clinician-rated scales like HAMD to ensure methodological robustness (Linardon et al., 2024). Moreover, the HAMD's reliance on in-person evaluations impose significant time and financial burdens. There is an increasing and urgent demand for depression assessment tools that are not only accurate and reliable, but also accessible and convenient to use.

Recent advances in large language models (LLMs) have opened new opportunities in natural language processing (Esteva et al., 2021). Studies have shown that LLMs exhibit potential to surpass human experts in medical knowledge-based tasks. For example, models like GPT-4

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https://doi.org/10.1016/j.ajp.2025.104707
Received 19 July 2025; Received in revised form 5 September 2025; Accepted 1 October 2025
Available online 1 October 2025

and Med-PaLM 2 demonstrate expert-level medical reasoning (Singhal et al., 2025). In terms of semantic parsing capabilities, models such as GatorTronGPT can accurately extract multi-source diagnostic information from electronic health records (Peng et al., 2023). In mental health assessment, LLMs have been used to analyze EHRs and develop an automated system to identify depressive symptoms with AUC of 0.91 (Pressman et al., 2024). In addition, LLMs can function as virtual patients, simulating realistic clinical interactions for research and training purposes. By responding to clinicians' inquiries based on preset conditions such as symptoms and medical history, these models help creating dynamic, reproducible patient interactions that support the development and evaluation of clinical tools (Cook et al., 2025).

Before the emergence of LLMs, traditional machine learning models for depression assessment relied on predefined, fixed-format questions, limiting their flexibility and adaptability to patient input (Shimamoto et al., 2024). As a result, these systems lacked the ability for dynamic, multi-turn dialogues with patients–making it difficult to clarify ambiguous responses or adjust conversational flow. On the other hand, LLMs also introduce certain limitations, most notably the risk of "hallucination"—the tendency to generate inaccurate or fabricated information due to their probabilistic and generative nature–which underscores the need of rigorous human oversight during clinical deployment (Wang et al., 2025). This may explain why most existing LLM-based depression assessment systems are based on retrospective datasets rather than real-time interaction with patients. For clinical applications, a practical effective way to mitigate hallucination risk is to constrain the model's behavior through carefully optimized prompts that enforce a structured and standardized interview process. (Omar et al., 2025).

To address these challenges, we introduce HAMLET (Hamilton Large-language-model Evaluation Tool), a novel framework that autonomously administers structured depression assessments using an LLM. Unlike prior models, HAMLET replicates the clinician-administered HAMD-17 protocol without professional involvement, while preserving its standardized evaluation logic. The goal of this study is to develop and validate a depression assessment tool that combines higher validity and diagnostic precision of clinician-rated scales with the accessibility, convenience, and low resource demands of self-report measures.

2. Methods

2.1. Participants

From February to June 2025, patients hospitalized in Beijing Huilongguan Hospital were recruited for the study. Inclusion criteria were: (1) diagnosis of Major Depressive Disorder (MDD) based on DSM-IV criteria, confirmed by the Structured Clinical Interview for DSM-IV (SCID); (2) age between 12 and 45 years; (3) proficiency in using digital devices (e.g., smartphones or computers); (4) provision of written informed consent by the patient or their legal guardian.

Exclusion criteria included: (1) comorbid organic brain disorders or serious, unstable physical conditions that could interfere with psychiatric treatment; (2) depression secondary to somatic illness, medication, or other psychiatric disorders; (3) severe adverse drug reactions requiring emergency intervention; (4) active suicidal ideation with imminent risk.

The study protocol was reviewed and approved by the Institutional Ethical Review Board of Peking Huilongguan Hospital and adhered to the Declaration of Helsinki.

2.2. Framework design

The core of the HAMLET assessment framework lies in the design and iterative refinement of prompt templates. The framework is designed to guide the language model through the structured logic of the HAMD-17, enabling accurate interpretation of patient responses and smooth transitions between assessment items, thereby ensuring the clinical validity of the overall evaluation process.

Initial prompt templates were developed by psychiatrists based on the official scoring criteria of the HAMD-17, with the aim to enable basic assessment functionality. However, early prompts often failed to guide the LLM appropriately. Drawing inspiration from the traditional training process of junior psychiatrists—who learn by observing expert assessments and receiving feedback—we implemented a simulation-based tuning workflow.

A "Patient Agent" was developed to serve as a virtual patient, emulating realistic responses based on pre-defined clinical profiles, including chief complaints, symptom descriptions, and disease trajectories. This agent interacted with an "Interview Agent" (i.e., the demo guided by current prompts) to simulate clinical assessments. Clinicians then reviewed these interactions, comparing the model's output and scoring against expert ratings, identifying discrepancies from expert ratings, and refining the prompt. Only after extensive simulation and refinement (Fig. 1, tuning phase) was HAMLET deployed for use with real patients.

HAMLET is built on the Qwen-Max (Bai et al., 2023), with a decoding temperature of 0.7 and a top-p value of 0.6. It operates in a purely text-based modality–accepting and generating text inputs and outputs–while patient interaction occurs via voice. The framework uses Edge-TTS (rany2, 2025) for converting the model's text output into speech, and Belle-whisper-large-v3 (BELLEGroup, 2023) for converting patient speech input into text. The transcribed patient response is passed to the evaluation framework (Fig. 1, Experimental Phase).

Each HAMD-17 items–or thematically related group of items (e.g., sleep disturbances or anxiety)– is associated with a tailored prompt. The LLM is instructed to output its evaluation in JSON format after each interaction to ensure accuracy and automation. Once a score is generated and stored, the system automatically proceeds to the next item, ensuring complete and streamlined coverage of the entire scale. All prompt transitions and scoring logic are handled in the background, allowing the patient to experience a seamless, natural-language interaction with an intelligent agent.

The full codebase for the HAMLET framework—including both tuning and experimental phases—is available open-source on GitHub (https://github.com/Enthoes1/HAMD). The specific prompt templates used in the assessment phase are not publicly released. The supplementary material provides a representative example from one of the HAMD items to illustrate the structure and logic of our approach.

2.3. Experimental procedure

A psychiatrist oversaw each patient's interaction with HAMLET to ensure safety and prevent the generation of harmful or misleading content by the LLM. No such adverse outputs were observed during the study. The supervising clinician could monitor the ongoing dialogue but had no access to the LLM's scoring results, which were stored in the backend and hidden from the patients and clinicians during the session.

Following the HAMLET assessment, each patient completed the PHQ-9 (Kroenke et al., 2001). To prevent bias introduced by prior evaluation–such as increased patient awareness of specific symptoms–the clinician-administered HAMD was conducted only after the HAMLET and PHQ-9 assessments. This sequencing ensured that the LLM operated independently, without the potential influence of earlier clinical questioning.

At the end of the session, patients were asked to rate their satisfaction with the HAMLET system on a three-point scale: dissatisfied, neutral, or satisfied. They were also asked to indicate their preferred assessment method among HAMLET, clinician-rated HAMD, and PHQ-9.

2.4. Data analysis

All statistical analyses and plots were performed using R version

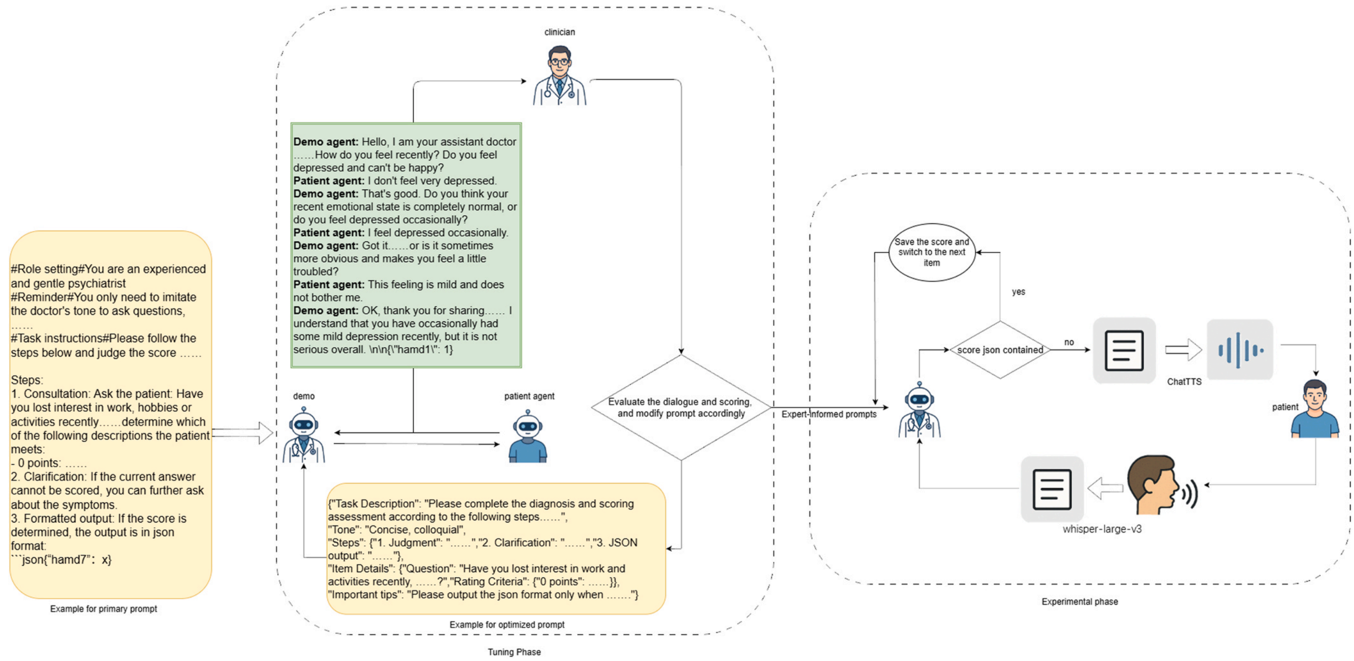


Fig. 1. Development framework of HAMLET.

4.4.2. Internal consistency of HAMLET was evaluated using Cronbach's alpha. A paired *t*-test was used to evaluate whether a systematic bias existed between HAMLET and HAMD. Given that HAMLET follows the same structure and scoring logic as the HAMD, it was treated as a new rater. Inter-rater agreement between HAMLET and clinicians was assessed using the Intraclass Correlation Coefficients (ICC) and Bland–Altman plots. The degree of item-level concordance was examined using both Gwet's AC2 and ICC.

To compare how strongly HAMLET and PHQ-9 scores correlated with clinician-rated HAMD scores, we used Steiger's Z-test. Incremental validity was evaluated through changes in adjusted R^2 values, assessing the extent to which HAMLET added explanatory power beyond that provided by the PHQ-9.

3. Results

A total of 60 patients diagnosed with major depressive disorder were enrolled in the study, comprising 24 males and 36 females, with a mean age of 25.3 years (SD = 10.3; range: 13–45). All participants completed the HAMLET assessment. The average duration of each HAMLET session was approximately 818.66 s (SD = 337.74).

The internal consistency of HAMLET scores was high, with a raw Cronbach's α of 0.822 and a standardized α of 0.818. Fig. 2a illustrates the distributional agreement between HAMLET and clinician-rated HAMD scores, with data points symmetrically distributed around the ideal agreement line (*y* = *x*). A paired *t*-test (t = –1.859, p = 0.07) indicated no significant systematic bias between the two methods (Fig. 2b).

To assess inter-rater agreement between HAMLET and clinician-rated HAMD scores, we computed the intraclass correlation coefficient (ICC) using a two-way mixed-effects model with single measurement (ICC (3,1)). The resulting ICC was 0.911 (95 % CI: 0.855–0.946), indicating strong agreement.

A Bland–Altman plot was constructed in Fig. 3a. While the average difference between HAMLET and clinician HAMD scores appeared minimal (mean difference = 0.72; 95 % LoA: −5.14 to 6.57), simple linear regression revealed a significant association between the mean of the two scores and their difference (β = 0.137, p = 0.01), indicating proportional bias. An adjusted Bland–Altman plot was subsequently

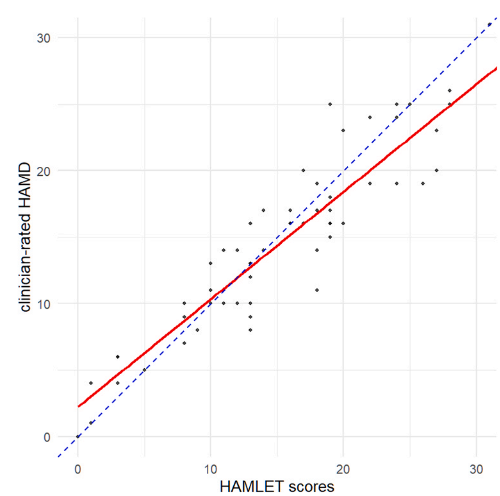


Fig. 2a. HAMLET and clinician-rated HAMD regression.
The blue dashed line represents y = x, indicating the ideal fit. The red solid line shows the regression line between HAMLET and HAMD scores.

generated (Fig. 3b) to account for this trend.

Item-level agreement between HAMLET and HAMD was further evaluated using ICC (Supplement Table 1). Most items demonstrated moderate to high consistency, except for Items 14 and 17, which showed relatively low ICC values (Supplement figure 1). This discrepancy may be partially attributed to the high proportion of zero scores, resulting in limited variance and biased ICC estimates (Mehta et al., 2018). To address this limitation and provide a complementary measure of agreement, Gwet's AC2 was calculated for each item, as shown in Fig. 4. All items exceeded the reference threshold of 0.60, suggesting at least moderate inter-rater agreement across the board.

Both HAMLET (r = 0.92) and PHQ-9 (r = 0.79) scores were strongly correlated with clinician-rated HAMD scores. To test whether HAMLET's correlation with HAMD was significantly stronger than that of the PHQ-9, Steiger's Z-test was conducted. The result (z = 3.798, p < 0.001; Fig. 5a), confirmed that HAMLET had a significantly stronger association with clinician-rated HAMD scores than the PHQ-9.

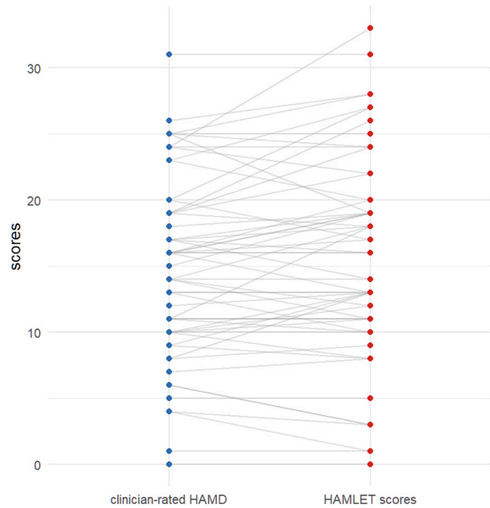


Fig. 2b. Paired-t test between HAMLET and HAMD.
No significant differences was found (t = –1.859, p = 0.07).

To assess incremental validity, stepwise linear regression was conducted. The change in adjusted R^2 was 0.252, indicating a significant improvement in explained variance (Fig. 5b). This suggests that HAMLET explains additional variance in HAMD scores beyond that accounted for by PHQ-9, demonstrating superior incremental validity.

As shown in Fig. 6a, the majority of patients reported being either moderately satisfied or satisfied with the HAMLET assessment, with only a minority expressing dissatisfaction. However, when asked to compare all three evaluation methods, patients demonstrated a clear preference for assessments conducted by human clinicians. Preferences between AI-based and self-report evaluations were nearly evenly split (Fig. 6b).

4. Discussion

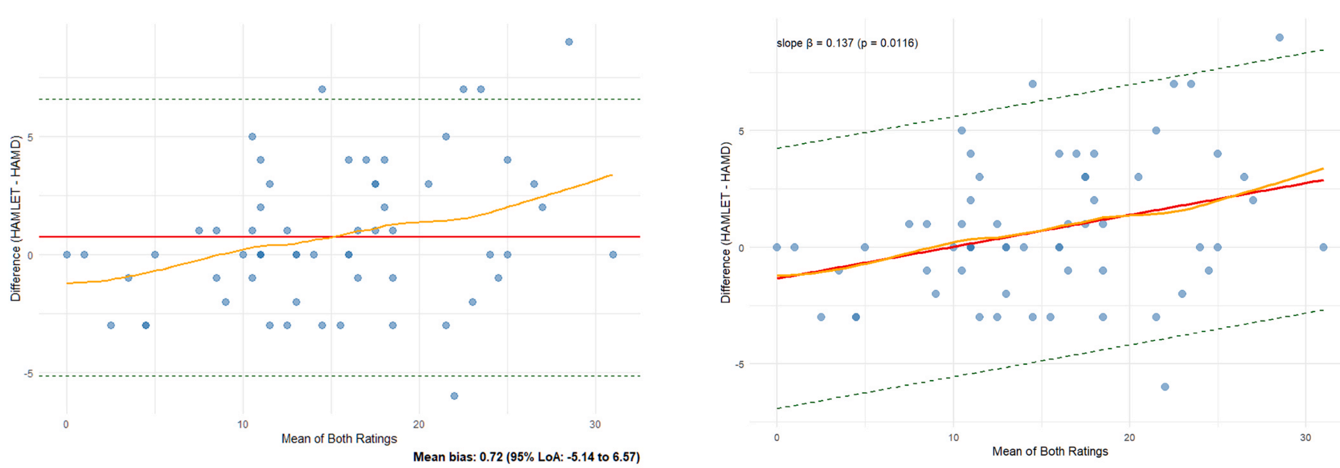
Recent years have seen a surge of interest in applying LLMs to clinical assessment and diagnosis. Prior studies have examined their effectiveness in answering treatment-related questions (Mejia et al., 2024), analyzing cognitive decline from electronic medical records (Du et al., 2024), recognizing synthetically written obsessive-compulsive scenarios (Kim et al., 2024), and predicting PHQ-9 scores based on existing medical texts (McCoy et al., 2025). However, most of these investigations have relied on either artificially generated data or preexisting clinical records, rather than real-time interactions with actual patients. Synthetic datasets cannot fully replicate the linguistic diversity and contextual nuances encountered in real clinical settings. Moreover, retrospective data are inherently shaped by clinician's judgment, which LLMs may simply reproduce, rather than reason through independently.

In contrast, our study advances the field by deploying an LLM in real-time to interact with patients, using carefully designed prompting and structured workflows. The direct engagement with real patients represents a significant step toward validating the feasibility and clinical utility of LLMs in psychiatric assessment.

The ICC between HAMLET and clinician-rated HAMD scores was 0.911, indicating strong overall consistency. However, a correlation between the differences and the mean values was revealed. The proportional bias observed in the Bland–Altman analysis suggests that HAMLET may overestimate symptom severity in patients with higher HAMD scores. One possible explanation is that severely depressed patients often display more pronounced negative verbal expressions and symptom features, which may be disproportionately weighted in the LLM's training data. As a result, the model might become "over-sensitive" to these signals. By applying regression-based limits of agreement, we explicitly accounted for this bias and provided a more accurate characterization of the agreement pattern. Importantly, despite this trend at the higher end of the scale, the overall agreement with clinician ratings remained substantial, as also supported by the ICC results.

At the item level, Gwet's AC2 values exceeded 0.60 across all items, suggesting moderate to excellent agreement. Items assessing measurable behaviors (e.g., weight loss, hypochondriasis, suicidal ideation) achieved particularly high AC2 values. This pattern aligns with LLMs' strength in extracting unambiguous, fact-based data via structured interview, where subjective interpretation is minimal.

However, certain items—particularly those related to sexual symptoms, anhedonia, and mood—showed lower agreement between the AI-based and clinician assessments. This discrepancy may stem from the nature of human–AI interaction. Knowing they were speaking to an AI, some patients may have responded differently to sensitive topics than they would in a face-to-face interview (Morris et al., 2018). However, as we did not collect patient feedback for each items, this explanation remains speculative and requires further validation. Additionally, for ethical reasons, the sexual symptoms item was automatically scored as 0 (not applicable) in patients under 18. This likely contributed to



(a) (b)

Fig. 3. Bland-Altman Plot and adjusted analysis comparing HAMLET and HAMD scores.
a. Solid red line represents mean bias. The two dashed lines indicate the upper and lower limits of the 95% LOA. The orange smooth curve represents the trend of the mean and difference. b. Regression-adjusted plot accounting for proportional bias. The slope (β) and p-value are annotated. Regardless of the original analysis or after correction, most of the points fall within the 95 % LoA in both Bland-Altman Plot and after adjusting.

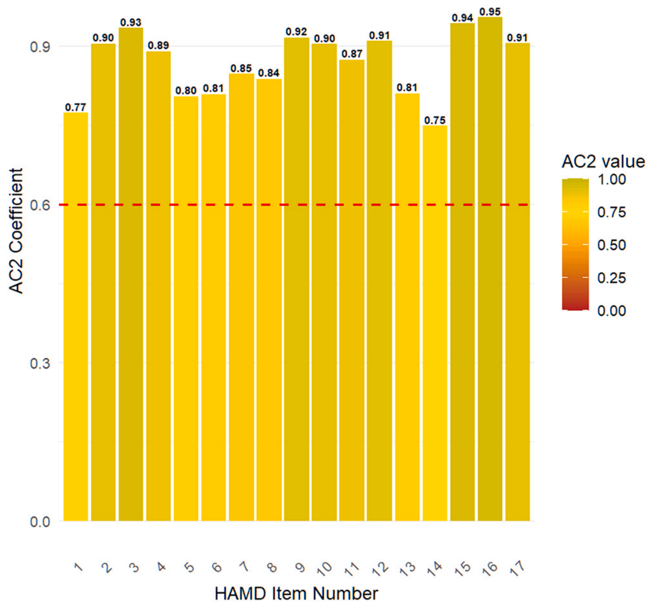


Fig. 4. Item-Level Agreement(Gwet's AC2) between HAMLET and HAMD. Red dashed line indicates good agreement threshold (AC2 ≥0.6).

reduced variance and lower item-level agreement. Nevertheless, it is worth noting that the Gwet's AC2 values for these items still exceeded 0.70, indicating acceptable reliability overall.

Our findings showed that HAMLET had a stronger correlation with HAMD scores than the PHQ-9. The PHQ-9, a widely validated self-report tool (Ma et al., 2021b), is valued for its simplicity and ease of use and is often used as a substitute for HAMD17. Despite its high correlation in our study ($r = 0.79$), self-report measures are susceptible to bias, potentially influenced by individual psychological traits (Ma et al., 2021a). Some studies have also suggested that free-text responses may offer a more accurate assessment of depressive symptoms than fixed-response formats, with many patients preferring open-ended questions (Sikström et al., 2023). Compared to the PHQ-9, HAMLET achieved stronger correlation with HAMD scores while maintaining a comparable administration time and requiring less human resource involvement.

As an exploratory study, the HAMLET framework demonstrates the potential of leveraging LLMs in assessing depressive symptoms using the HAMD criteria. However, several limitations remain and warrant consideration in future research.

To optimize assessment accuracy, HAMLET currently adopts a modular prompt-based approach, where each item or symptom cluster is addressed in a separate interaction. This approach compensates for a known limitation of LLMs in processing extended contexts—specifically the tendency to "forget" earlier information in long dialogues– due to context-length constraints. As dialogue progresses and becomes more complex, the model's ability to recall and integrate earlier information diminishes significantly (Liang et al., 2024). Furthermore, densely layered prompts can undermine instruction-following accuracy as the model struggles to allocate attention evenly across long contextual inputs, increasing the risk of omitting critical information (Zhang et al., 2024), impairing overall task performance (Kiyomaru et al., 2024). In this study, we selected Qwen-Max as the base model. Future iterations may benefit from more advanced LLMs–such as GPT-4.5, Qwen3 or Claude 3.7–which offer improved contextual memory and robust prompt adherence. As LLMs evolve, assessments may no longer need to be segmented, enabling more natural, continuous interactions.

Another limitation relates to the generative nature of LLMs, which can lead to outputs instability (Huang et al., 2025) Future efforts could explore parameter-efficient fine-tuning methods like LoRA (Taylor et al., 2024) to enhance psychiatric assessment accuracy in more complex conversational context. However, some studies caution that continued fine-tuning may paradoxically reduce instruction-following accuracy, thereby impairing task performance (Jiang et al., 2024). For this reason, we prioritized iterative prompt engineering over fine-tuning to optimize performance. Nonetheless, domain-specific fine-tuning could be valuable in the future, particularly for improving the naturalness of spoken interactions. While HAMLET currently engages patients via voice-based interaction, scoring decisions are solely based on textual outputs. Future frameworks could incorporate multimodal data—such as vocal tone or facial expressions—to improve precision and robustness.

In summary, this study developed and validated an LLM-based framework for evaluating HAMD scores, demonstrating the feasibility, validity, and accuracy of using LLMs to perform structured psychiatric assessments. The HAMLET framework showed strong agreement with clinician-rated HAMD scores and outperformed PHQ-9 in correlation and incremental validity. These findings support the potential of LLMs to assist or even autonomously conduct standardized psychiatric evaluations.

Traditionally, the development of such frameworks required close collaboration between clinicians and programmers. However, with the emergence of LLMs capable of generating code, this framework was designed and implemented entirely by clinicians. We named it HAMLET—Hamilton Large-language-model Evaluation Tool—as a deliberate nod to Shakespeare's enduring existential question: "To be, or not to

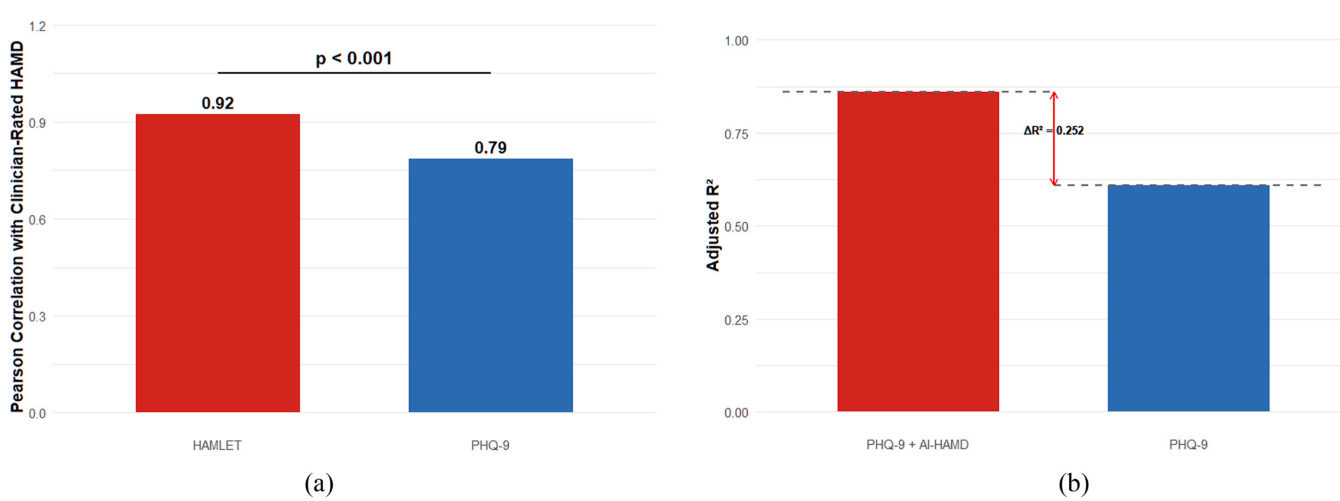


Fig. 5. Comparative Correlation and Incremental Validity of HAMLET vs PHQ-9 Relative to Clinician HAMD Ratings.

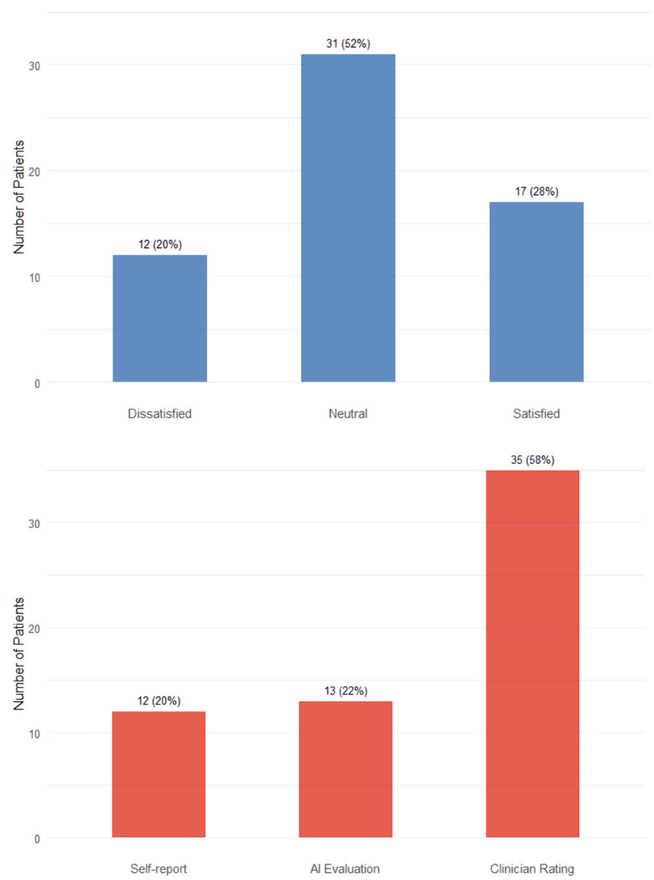


Fig. 6. Comparative Patient Satisfaction with HAMLET vs Traditional Assessment Methods.

be?" As machines begin to interpret the human mind, we must continuously ask: does the rise of instrumental rationality through AI preserve our capacity for human understanding and empathy? Can these tools truly replace clinician judgment, or do they best serve by augmenting it?

CRediT authorship contribution statement

Haitao Chen: Validation. **Shuping Tan:** Project administration, Funding acquisition, Conceptualization. **Wen Xin:** Writing – original draft, Visualization, Software, Methodology, Formal analysis. **Huailiang Yi:** Validation, Data curation. **Jiaqi Song:** Resources, Project administration. **Zhanxiao Tian:** Methodology.

Funding

This study was supported by the Capital Funds for Health Improvement and Research (2022-1G-2132), Municipal High-Level Public Health Talent Project of Beijing (Academic Lead-02-03), Beijing Natural Science Foundation (7242077), Beijing Municipal Science & Technology Commission Grant (Z191100006619104), and Beijing Hospitals Authority Ascent Plan (DFL20192001).

Declaration of Competing Interest

The authors declare that they have no competing financial interests or personal relationships that could be perceived to have influenced the work reported in this paper.

Acknowledgements

We would like to thank Editage (www.editage.cn) for English language editing.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ajp.2025.104707.

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